

# JUNQI LU

[GitHub](#) | [Google Scholar](#) | [Personal Blog](#)

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Beijing, China

## EDUCATION

Beijing Institute of Technology

Beijing, China

B.E. in Computer Science (Second Bachelor's Degree)

Sep. 2024 – Jun. 2026 (Expected)

- **Ranking:** Ranked 1st in the program.

B.S. in Mathematics and Applied Mathematics

Oct. 2020 – Jun. 2024

- **Thesis Grade:** Excellent (Top tier of graduation projects).
- **Focus:** Rigorous training in Real Analysis, Algebra, and Probability Theory.

## PUBLICATIONS

[1] **Junqi Lu**, Bosen Liu, Cuicui Pei, Qingan Qiu\*, and Li Yang\*. Learning to optimize termination decisions under hybrid uncertainty of system lifetime and task duration. *Computers & Industrial Engineering*, 2025. DOI: [10.1016/j.cie.2025.111208](https://doi.org/10.1016/j.cie.2025.111208) (Published, IF=6.5, JCR Q1)

[2] **Junqi Lu**, Ruixiang Sun, Xin Li\*, et al. Escaping the Variance Trap: Bisimulation Saddle-Point Optimization for Robust Behavioral Representation Learning. (Manuscript Under Review)

[3] **Junqi Lu**, Qingan Qiu\*, Yike Zhang. On the Optimality of Myopic Policies for Condition-based Termination under Parameter Uncertainty. (Working Paper)

## RESEARCH EXPERIENCE

Deep Reinforcement Learning & State Representation

Jul. 2025 – Present

Research Assistant | Advisor: [Prof. Xin Li](#) | [1029 DRL Lab](#) | Beijing Institute of Technology

- **Theoretical Analysis:** Identified the "Variance Trap" in metric learning. Proved via **Stochastic Differential Equations (SDEs)** that conventional regression objectives force representation collapse in stochastic environments.
- **Methodology (BSPO):** Proposed a primal-dual framework (*Bisimulation Saddle-Point Optimization*) with an **Auxiliary Network** to decouple structural error from transition noise, eliminating estimation bias.
- **Evaluation:** Achieved SOTA performance on **DeepMind Control Suite** under high-noise and visual distractions, significantly outperforming baselines (MICO, SimSR).
- **Outcome:** First-author manuscript completed and submitted for peer review.

Reliability Engineering & Stochastic Control

Jun. 2023 – Present

Research Assistant | Advisor: [Prof. Qingan Qiu](#) | Beijing Institute of Technology

- **Mission Abort Framework:** Proposed an **MDP** framework to balance task success probability and system safety under hybrid uncertainty. Formulated a discrete optimization method to derive optimal termination thresholds.
- **Theoretical Extension:** Addressed epistemic parameter uncertainty via a **Bayesian POMDP** approach. Rigorously proved that the optimal control policy structurally degenerates into a **Myopic Bang-Bang Policy** in safety-critical regimes, eliminating the need for complex look-ahead planning.
- **Outcomes:** Published one first-author paper in **CAIE (Q1, IF=6.5)** and preparing a second manuscript.

SELECTED PROJECTS

- Nitrogen Cycling Dynamics Model (MCM-ICM 2025 Finalist)[GitHub: 100+ Stars]
- **Modeling:** Developed a dynamic ecosystem model using modified **Lotka-Volterra equations** to simulate nitrogen flow across forest and agricultural food webs.
  - **Open Science:** Released a fully reproducible solution repository with comprehensive documentation and tutorials. Garnered **100+ GitHub Stars**, widely recognized by the modeling community for its code quality and educational value.
  - **Award:** Selected as a **Finalist Winner (Top 2%)** out of 27,456 teams globally.

- High-Performance Concurrent BBS & Chat System (C++)[GitHub Code]
- **Architecture:** Engineered a high-performance, event-driven server on Linux using **Epoll** (I/O Multiplexing) to efficiently manage massive concurrent connections with minimal resource overhead.
  - **Protocol Design:** Designed a custom **Binary Application-Layer Protocol** (Header + Payload) to ensure reliable TCP message framing and implemented a secure handshake mechanism for file transmission.
  - **System Engineering:** Adhered to strict modular design (User/Chat/File Managers) and implemented data persistence logic; realized distinct features including group broadcasting and peer-to-peer file relay.

HONORS & AWARDS

- **Finalist (Top 2%)**, Interdisciplinary Contest in Modeling (ICM)Feb. 2025
- **Outstanding Student Award** (Top 10% university-wide), BITNov. 2025
- **Academic Excellence Scholarship** (Rank 1st in CS Program), BITOct. 2025
- **University "Excellent Thesis" Prize**, BITJun. 2024
- **Meritorious Winner (Top 9%)**, Mathematical Contest in Modeling (MCM)Feb. 2023
- **Second Prize**, China Undergraduate Mathematical Contest in Modeling (CUMCM)Nov. 2022

RELEVANT COURSEWORK

- **Computer Science:** Data Structures and Algorithms, Operating Systems, Computer Architecture, Computer Networks, Object-Oriented Programming, Database Principles.
- **Mathematics:** Real Analysis, Functional Analysis, Abstract Algebra, Probability Theory, Mathematical Statistics, Partial Differential Equations (PDE), General Topology, Numerical Analysis.
- **Specialized Topics:** Reinforcement Learning (Graduate-Level), Machine Learning Fundamentals.

SKILLS

|                       |                                                                                                   |
|-----------------------|---------------------------------------------------------------------------------------------------|
| Programming Languages | Python (PyTorch, NumPy, Pandas), C++, SQL, MATLAB, Bash                                           |
| AI Workflow           | Chinese (Native), English (B2), German (A1)                                                       |
| Tools                 | Proficient in AI-assisted coding (VS Code + Copilot/Cursor) for efficient development.            |
| Outreach              | Git, LaTeX, Makefiles, Lean 4 (Formal Proof Verification)                                         |
|                       | <b>Tech Blogger</b> ( <a href="#">CNBlogs</a> ) with 20K+ views (20+ articles on DSA & LeetCode). |

INTERESTS

- **Marathon Running:** Committed long-distance runner with an annual mileage exceeding **1,000 km** for three consecutive years. Half-Marathon Personal Best: **1:41:34**.
- **Music:** Lead guitarist and bassist for university rock bands. Passionate about classic rock arrangement and performance; active participant in campus music festivals.